Evaluation of ECMWF S2S Forecasts over Ireland for Multi-Hazard Applications

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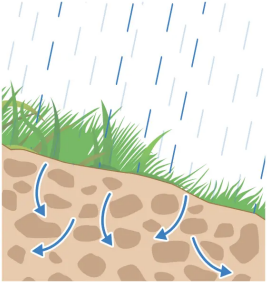
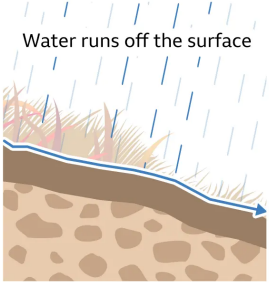
What Happens When Drought Becomes Flood?

Normal conditions



Soil absorbs water like a sponge

Drought conditions



Hard layer of soil repels water

Adapted from BBC Research (2022)

$\sim$25%

of floods globally preceded by drought

Matanó et al. (2024)

up to 8$\times$

economic & humanitarian impact

Worou & Messori (2025)

$\sim$2$\times$

compound events by mid-century

Steensen et al. (2025)

Has Anyone Tried to Forecast This?

Research landscape: drought–flood prediction

	Retrospective	Forecast	Projection
Hazards treated independently	Drought and flood inventories (SSI, HANZE, EM-DAT, ...)	Operational flood and drought products (GloFAS, EFAS, EDO, ...)	Single-hazard climate projections (CMIP6, UKCP18, CORDEX, ...)
Hazards treated as connected	**57 papers** DFAA Whiplash D-to-F UK hydro-haz. Weather whip.	★	Steensen et al. (2025) Tan et al. (2023) Collet et al. (2018) Formetta et al. (2019) Visser-Quinn et al. (2019)

Is This Forecastable?

Trigger is predictable

53%

of drought endings driven by Atlantic IVT

Parry et al. (2023)

→

Where they translate to ground

1 in 5

reach the ground

>2 in 5 in similar catchments

Brunner et al. (2025)

→

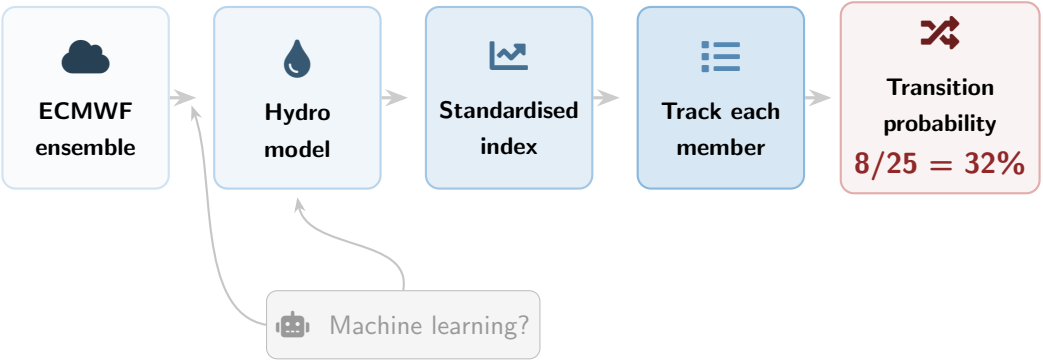
Open questions

?!

Skill?

Define transition?

What's My Approach?



Takeaways

Transitions cause worse impacts

Nobody has forecast them yet

Ireland as my testbed

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References

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